

18-427/727: Board-Level RF Systems for IoT

LAB05: Antenna Array Simulations

Writeup Due: Friday, November 14th, 2025

How to submit labs:

Create a word document or PDF that answers the questions presented below. Include any screenshots or figures as requested. **Please provide a descriptive caption for each figure!**

Question:	1	2	Total
Points:	50	50	100
Score:			

Lab Outline

This lab will explore antenna arrays using MATLAB's Antenna Toolbox, the simulation tool that you used in Lab 2. You will learn how to steer the radiation pattern in a particular direction. This will help to develop your understanding for next week's lab session, where we will perform beam steering with a physical implementation of a dipole antenna array and a KrakenSDR.

1. Linear Dipole Arrays
2. Patch Arrays

Equipment Required

- Laptop - to run MATLAB

1. Linear Array

Let's get reacquainted with the MATLAB Antenna Toolbox by designing and simulating a *linear array* of half-wavelength dipole antennas. We will design antennas for use in the 915 MHz ISM band. Use the following code, setting `dipole_length` to the appropriate value to create the `dipole` object and visualize its patterns at $f = 915$ MHz.

```
1 dipole = dipoleCylindrical('Radius', 1e-3, 'Length', dipole_length);
2
3 figure;
4 patternAzimuth(dipole, 915e6); % show the azimuth pattern
5
6 figure;
7 patternElevation(dipole, 915e6); % show the elevation pattern
8
9 figure;
10 pattern(dipole, 915e6); % show the radiation pattern at 915 MHz
```

That should be review from Lab 2. Now let's lay out an array. Create a `linearArray` object and set it to use $N=4$ of your `dipole` objects with $d = \lambda/2$ element spacing. `layout()` visualizes your array layout.

```
1 la = linearArray;
2 la.NumElements = 4;
3 la.Element = dipole;
4 la.ElementSpacing = ; % set this to lambda/2
5 layout(la);
```

Once you've made your `linearArray` object `la`, visualize its patterns at $f = 915$ MHz.

```
1 figure;
2 patternAzimuth(la, 915e6); % show the azimuth pattern
3
4 figure;
5 patternElevation(la, 915e6); % show the elevation pattern
6
7 figure;
8 pattern(la, 915e6); % show the radiation pattern at 915 MHz
```

10 pts

1.1 Provide each of the following plots of your $N = 4$ linear array:

1. Layout
2. 3D pattern
3. Elevation pattern
4. Azimuth pattern

5 pts

1.2 What is the maximum directivity of this antenna array, and how does it compare to the single dipole's maximum directivity?

5 pts

1.3 What is the 3dB beamwidth of the main lobe in your azimuth pattern in degrees?

Hint: You can call `directivity = patternAzimuth(la,915e6);` to store the directivities in a linear array. Plotting these over a rectangular plot might make it easier to compare lobes than on the polar plot.

5 pts

1.4 What is the ratio between the main lobe to the strongest sidelobe in dB for the azimuth pattern?

10 pts

1.5 Now, generate azimuth patterns for $N = 8$ and $N = 16$ linear arrays to compare to your patterns for $N = 4$. Then, include a table with the following data:

1. Maximum directivity vs N
2. 3dB beam width vs N
3. Azimuth pattern's main lobe to sidelobe ratio vs N

10 pts

1.6 Steer the antenna beam to $\phi = 45^\circ$ azimuth for your $N = 16$ array. You can apply a series of phase shifts to the linear array object `la` with the following line of code:

```
la.PhaseShift = phases;
```

where `phases` is an $N \times 1$ vector of phase shifts to apply to each antenna element in degrees.

To steer a beam towards ϕ in azimuth, you should apply this *successive* phase shift ψ to each antenna in the array:

$$\psi = \frac{2\pi}{\lambda} d \cos(\phi)$$

Note that d is the distance between two antennas, which should be a half-wavelength. Also, note that the `PhaseShift` property in MATLAB expects a scalar or vector in degrees.

When you have steered the beam to your liking, please provide the following:

1. Azimuth pattern of the beam steered to $\phi = 45^\circ$
2. The `phases` vector you used

5 pts

1.7 You generated these patterns simply by shifting signal phases to each element in the array (i.e. applying a shift $w_n = e^{j\psi_n}$). What could you gain by modulating amplitude as well as phase (i.e. applying a complex weight $w_n = A_n e^{j\psi_n}$) to each element in the array? What is this technique called?

2. Patch Array

Now create a microstrip patch antenna with a center frequency of 915 MHz with the following command:

```
1| patch = design(patchMicrostrip, 915e6);
```

Then, using the `linearArray` function, establish a 4x1 linear array with $d = \lambda$ element spacing.

10 pts

2.1 Provide each of the following plots of your $N = 4$ linear array:

1. Layout
2. 3D pattern
3. Elevation pattern
4. Azimuth pattern

10 pts

2.2 Adjust the phase shifts to the patches to create two different beams that might support MIMO, and provide plots of the Elevation pattern of each beam. Please justify your choices and share the phase shifts used to create each pattern.

10 pts

2.3 Use the `rectangularArray` function to create an 2x2 rectangular patch array, with $d = \lambda$ spacing between rows and columns of elements. Provide the plots of the following:

1. Layout
2. 3D pattern
3. Elevation pattern
4. Azimuth pattern

10 pts

2.4 Adjust the phase to steer the beam to 60 degrees elevation (and 0 degrees azimuth). Use the MATLAB `phaseShift()` function to calculate the necessary shifts! Then provide a plot of the Elevation pattern along with the phase shifts used.

10 pts

2.5 How do the patterns of these two patch arrays differ from one another? In what situation might you prefer the 4x1 linear array over the rectangular 2x2 array?

Reflections

Please complete our [post-lab survey](#) for 5 points of extra credit! Attach a photo of the form completion page to your lab submission.